Teaching Guidelines for

MS.Net Technologies

PGCP-AC February 2026

Duration: 120 hours (45 theory hours + 45 lab hours +30 Self-Learning hrs)

Objective: To acquire the knowledge of Microsoft.NET 8.

Prerequisites: Students are expected to know any OOP. They should have undergone the Web Programming module which includes HTML, CSS, JavaScript, JSON, and XML. Knowledge of any database is required.
Note: Training will be carried out on .Net8 using Visual Studio 2022

Evaluation: 100 marks
Weightage: CCEE– 40%, Lab exam – 40%, Internals – 20%

Text Book:
- C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals by Mark J. Price

References:
- C# 12 in a Nutshell: The Definitive Reference Book by Joseph Albahari

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(Note: Each Session is of 2 hours)

Session 1: : **(2T)**
Lecture:
Introduction to the .Net Framework
Intermediate Language (IL)
Assemblies and their structure, EXEs/DLLs
CLR and its functions
- JIT Compilation
- Memory Management
- Garbage Collection
- AppDomain Management
- Memory Management
- CLS, CTS
- Security

No Lab

Session 2: : (2T+ 3SL)
Lecture:
.Net Framework, .Net Core, Mono, Xamarin differences
Versions of the Framework
Managed and Unmanaged Code
Introduction to Visual Studio
Using ILDASM
No Lab

Session 3: : (2T+ 4L)
Lecture:
Console Applications and Class Libraries.Net Core
C# Basics
Project References, using
Classes
Data Types in .net and CTS equivalents
Methods
- Method Overloading
- Optional Parameters
- Named Parameters and Positional Parameters
- Using params
- Local functions

Properties
- get, set
- Readonly properties
- Using property accessors to create Readonly property

Constructors
Object Initializer
Destructors
Discussion on IDispose. To be implemented after interfaces
Lab:
Create a class that has Properties, Fields, Methods, Constructors (Trainer can specify any class of his choice, e.g. Student, Employee, etc)

Session 4: : (2T+ 4L+ 3SL)
Lecture:
Static Members of a Class
- Fields
- Methods
- Properties
- Constructors

Static Classes
Static local functions
Inheritance
- Access Specifiers
- Constructors in a hierarchy
- Overloading in derived class
- Hiding, using new
- override
- sealed methods
- Abstract Classes
- Abstract Methods
- Sealed Classes

Lab:
Create multiple classes that use Inheritance based concepts

Session 5: : (2T+ 2L)
Lecture:
Interfaces
- Implementing an interface

- Explicitly implementing an interface
- Inheritance in interfaces
- Default interface methods

Operator overloading

Lab:

Create and implement interfaces for the classes created in Lab 4

Implement IDisposable, IComparable

Session 6: : (2T+ 2L)

Lecture:

Reference and Value Types

Value Types

- struct
- enum

out and ref

nullable types

nullable reference types

?? and ??=

Working with Arrays (single, multidim, jagged), Array Class members

Indices and ranges

Indexers

Lab:

Lab based on array examples.

Also create an array of the class created in Lab 1.

Session 7: : (2T+ 2L+ 3SL)

Lecture:

Generic classes

Generic methods

Generic Constraints

Collections – generic and non-generic

Collection Examples based on ICollection, IList, IDictionary (both generic and non-generic)

Iterating collections using foreach

Using Tuples to pass multiple values to a function

Lab:

Lab based on collection examples.

Also create a collection of the class created in Lab 1.

Session 8: : (2T+ 2L)

Lecture:

Delegates

- Calling methods using delegates
- Uses of delegates
- Multicast delegates
- Action, Func, Predicate delegates

Anonymous methods

Lambdas

Lab:

Lab based on delegates examples.

Session 9: : (2T+ 2L+ 3SL)
Lecture:
Error Handling (Exceptions Handling)

- Checked & Unchecked Statements
- The try, catch, finally
- Dos & Don’ts of Exception Handling

User Defined Exception classes
Declaring and raising events
Handling events
Lab:
Lab based on exceptions and events examples.

Session 10: : (2T+ 2L+ 3SL)
Lecture:
Anonymous types
Extension methods
Partial classes
Partial methods
LINQ to objects
Writing LINQ queries
Deferred execution
LINQ methods
PLINQ
Lab:
Lab based on LINQ examples
Students to try tutorial for 101 LINQ Queries

Session 11: : (2T+ 3SL)
Lecture:
Creating a shared assembly
Creating Custom Attributes
Using Reflection to explore an Assembly
Using Reflection to load an Assembly dynamically
Files I/O and Streams

- Working with drivers, Directories, and Files
- Reading and Writing files

No Lab

Session 12: : (2T+ 4L)
Lecture:
Threading

- ThreadStart, ParameterizedThreadStart
- ThreadPool
- Synchronizing critical data using lock, Monitor and Interlocked

Working with Tasks

- Calling functions with and without return values
- Using async, await

Using the Task Parallel Library
Lab:
Threading related examples
Task related examples

Sessions 13-19: : (14T+ 14L+ 6SL)
Lecture:
Introduction to Asp.Net MVC CORE

- Architecture of an ASP .Net MVC application
- Understanding Folder structures and configuration files

Understanding Controllers and Action

- Create a controller
- How actions are invoked
- HttpGet , HttpPost , NoAction Attributes
- Running Action result.

Understanding Views & Models

- Creating Models & ViewModel
- Creating Razor Views
- HTML Helper Functions
- Understanding ViewBag
- Create a view using ViewBag
- Validation using Data Annotations
- Client side and server side validation
- Self validated model
- Creating Strongly Types Views
- Using Various Scaffold Templates
- CRUD operation using Model

MVC State Management

- ViewBag , TempData , Session , Application
- Cookies , QueryString

MVC Module

- Partial View
- Action Method and child action

Data Management with ADO.NET

- Microsoft.Data.SqlClient introduction
- Connection object, Command object, DataReader, DataAdapter, DataSet and DataTable.
- Asynchronous command Execution
- Asynchronous Connections

Understanding Routing & Request Life Cycle

- Routing Engine & Routing Table
- Understanding and configuring Routing Pattern in RouteConfig File
- Understanding 404 error and resource not found.
- Using Attributes Routing
- Understanding Request Life Cycle

Layouts, Bundle, Minification

- Creating Layout and using with associated views
- Understanding Bundling and Minification
- Using BundleConfig file
- Attaching css , js , bootstrap in bundles
- Custom Helper Function
- Asynchronous Actions
- Error Handling in MVC with Log Entry
- Filters and Custom Action Filter

MVC Security

- Using Authorize & Allow Anonymous attributes
- Implementing Forms Based Authentication

- Preventing Forgery Attack using AntiForgeryToken
- Preventing Cross Site Scripting Attack

Entity Framework

- Introduction to EF
- Different Approaches
- Using Code First Approach
- Using various Data Annotations
- Using Validation, Primary Key , Foreign Key etc
- Using Fluent APIs
- Database Migrations
- CRUD operation using EF

Developing MVC application using EF Code First Approach
Introduction to Razor Pages

Lab:
Lab exercise covering the concepts covered in the class

Session 20: : (2T+ 2L+ 3SL)
Lecture:
Localization in MVC (Demo Only)
Deploying ASP.NET MVC application (Demo only)
Web APIs

- Creating ASP.NET MVC Web API
- Configuring for CORS
- Different Verbs
- Consuming using a client
- Using Newtonsoft APIs
- Integrating Web API with React App
- Configuring CORS of React App and Web API
- Sending request from React App, processing at Web API and effecting the database

Lab :
Create a RESTful service using Web API.
Create a consumer.

Sessions 21 & 22: (5T+ 5L+ 3SL)
Lecture:
MVC integration with React

- Introduction to MVC and React
- Setting up the Project
- Integrating React with MVC Backend
  - Define Models
  - Implement Controllers
  - Use Views
- Data Management and State Handling
  - Establish data flow
  - Manage state
- Advanced Topics and Best Practices
  - Authentication and Authorization
  - Routing
  - Structuring React components

Lab:

Create a RESTful service using WEB API with React as the front end.
Create a consumer.